AI4MH: Governance Readiness Design

AI-Powered Behavioral Analysis for Suicide Prevention, Substance Use & Mental Health Crisis Detection

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Google Summer of Code 2026 — Contributor Selection Task
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GitHub: github.com/Swasti2004/gsoc2026-ai4mh-governance

AI4MH is a public health monitoring system designed to give health personnel an early warning layer for detecting emerging mental health and substance use crises before they appear in hospital admissions or overdose data. The 2025 GSoC prototype built the foundational pipeline: data ingestion, BERT-based risk classification, and sentiment analysis. This submission adds the governance layer on top of that foundation. The system is not an autonomous decision-maker; it is a technical tool that surfaces all relevant evidence to a human reviewer, who retains full authority over every consequential decision. Through audit logging, analyst overrides feed back into the system over time, making it progressively more reliable. The governance design also accounts for real world conditions that would compromise signal reliability-bot amplification, media-driven spikes, and rural underrepresentation ensuring the system behaves responsibly across the full diversity of communities it serves.

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1. Crisis Signal Design

1.1 Framework Philosophy

The three signal components — sentiment intensity, volume spike, and geographic clustering ,are not treated as equal inputs to a simple weighted average. Sentiment is the anchor signal because it directly reflects the language people are using to express distress; an extreme sentiment score carries sufficient weight to stand alone without corroboration. Volume is a confidence modifier, not an independent signal , a spike in post volume means nothing until you interrogate why it occurred. A media event covering a local crisis can drive hundreds of posts that discuss the news rather than express personal distress, making raw volume deeply misleading without context. Geographic clustering serves as a corroborating modifier; when neighboring counties show similar distress patterns, confidence in the signal increases. Critically, the absence of geographic clustering does not invalidate a signal; a rural county with limited social media activity can still be experiencing a genuine crisis, and the system accounts for this rather than penalizing it.

1.2 The Three Signal Components

Conditional weighting logic: If sentiment alone is extreme (BERT risk score > 0.85), the CSS treats it as a sufficient signal regardless of volume or geography because a genuine crisis doesn't stop being a crisis because fewer people are posting. For moderate sentiment (0.45–0.85), volume and geographic clustering are required to corroborate before escalation.

Component	Role	Source in Prototype	Key vulnerability
Sentiment Intensity	Anchor signal — primary indicator of distress language	BERT classifier + VADER heuristics	Coded or slang language may be missed
Volume Spike	Confidence modifier — corroborates sentiment	Post count vs. 30-day rolling baseline	Bot amplification, media-driven spikes
Geographic Clustering	Corroborating modifier — regional pattern confirmation	Neighboring county comparison	Rural sparsity, uneven platform usage

1.3 Minimum Sample Size Threshold

A hard gate is applied before scoring begins. Urban counties require a minimum of 30 posts within the 72-hour window. Rural counties (RUCC codes 7–9) use a reduced threshold of 15 posts, reflecting structurally lower social media activity , not lower crisis risk. Counties below their respective threshold are never silenced; they are routed to a dedicated sparse-data human review queue with all available evidence presented and explicit uncertainty labeling applied.

1.4 Smoothing — Exponential Weighted Moving Average (EWMA)

Daily sentiment scores and post volumes are smoothed using EWMA before scoring. The intuition is straightforward: previous days are taken into account, but weighted so they do not overpower the current signal. A single-day spike caused by a local news story is dampened; a genuine multi-day escalation trend is preserved and amplified. EWMA was chosen over a simple rolling mean because it responds faster to genuine trend changes while being more resistant to one-off noise events.

1.5 Confidence and Uncertainty Estimate

Uncertainty is communicated to the human reviewer simultaneously through three channels, because a public health officer needs uncertainty to be impossible to miss and impossible to misinterpret:

- **Percentage confidence** — derived from the degree of agreement between the three signal components. High sentiment + high volume + geographic confirmation yields high confidence. Sentiment alone yields lower confidence with a wider uncertainty band.
- **Plain-language explanation** — auto-generated one-line description of what is driving the signal, e.g. *"Strong signal — high sentiment across 3 neighboring counties"* or *"Weak signal — 14 posts only, rural sparse data flag applied."*
- **Visual indicator** — color-coded tier display on the dashboard with confidence band shown as a range, not a single number, to prevent false precision.

1.6 Pseudocode — Crisis Signal Scoring Logic

```
function evaluate_county(county, window=72hrs):

  posts = collect_posts(county, window)

  # Step 1: Sample size gate
  threshold = 15 if rural(county) else 30
  if len(posts) < threshold:
      route_to sparse_data_queue with uncertainty label
      stop

  # Step 2: Clean the signal
  flag_and_discount_bots(posts)         # velocity + cosine dedup filter
  flag_and_discount_media_spike(posts)  # timing vs news registry + language register
  smooth_scores = apply_EWMA(posts, previous_days)

  # Step 3: Score each component
  S = bert_risk_score(posts)            # BERT classifier output [0,1]
  V = volume_z_score(posts, baseline)   # Z-score vs 30-day rolling mean, capped at 4.0
  G = geographic_clustering(county)     # Moran's I across neighboring counties

  # Step 4: Conditional weighting
  if S > 0.85:                          # extreme sentiment — anchor is sufficient
      CSS = S
  else:                                 # moderate sentiment — corroboration required
      V_adjusted = V * 0.5 if bot_flag or media_flag else V
      CSS = 0.45*S + 0.35*V_adjusted + 0.20*G

  # Step 5: Confidence estimate
  component_agreement = measure_agreement(S, V, G)
  confidence = {
      percentage: component_agreement,
      plain_language: generate_explanation(S, V, G, flags),
      visual: assign_color_tier(CSS, confidence_band)
  }

  # Step 6: Escalation routing
  if CSS >= 0.75:  escalate -> senior_reviewer (immediate)
  if CSS >= 0.50:  escalate -> oncall_reviewer (within 4hrs)
  if CSS >= 0.30:  log -> daily_review_queue
  else:            log -> noise_archive

  # Step 7: Write audit record
  audit_log.append(county, window, posts_n, S, V, G, CSS,
                   flags, confidence, escalation_tier)
```

2. Governance & Risk Controls

The thread connecting every governance control in this section is a single principle: the system must never hide its own uncertainty or limitations from the human reviewer. Every mitigation below — discounting, flagging, transparency about bias, audit logging , is a different expression of that same commitment.

2.1 Bot Amplification & Coordinated Activity

Bot activity and coordinated inauthentic behavior inflate volume without reflecting genuine community distress. The system detects this through behavioral patterns specific to Reddit (account age, karma score, posting velocity) , the platform used in the 2025 prototype:

- **Velocity filter:** Accounts posting more than 10 crisis-language messages within 24 hours in the same county are flagged. Their posts are included but their volume contribution is discounted by 50%.
- **Near-duplicate clustering:** Posts with TF-IDF cosine similarity above 0.92 within a 6-hour window are identified as coordinated. The cluster contributes one post-weight to volume, not N.
- **Bot risk flag:** If volume Z-score exceeds 3.0 but fewer than 20 unique account IDs contribute, a bot_risk flag is attached to the CSS record and surfaced prominently to the reviewer.

2.2 Media-Driven Spikes

High-profile news coverage drives reactive discussion that resembles distress escalation in volume but not in genuine community crisis. Detection uses two complementary methods:

- **Temporal pattern detection:** A manually maintained media event registry (updated by analysts within 24 hours of major coverage) is cross-referenced against volume spikes. Alignment within a ± 48-hour window triggers a media_context flag.
- **Linguistic register classification:** The 2025 BERT model distinguishes reportative language (*"did you hear about..."*) from first-person distress (*"I cannot go on"*). Media-reactive posts receive a 0.6× volume weight multiplier. Critically, the flag is surfaced to the reviewer — the algorithm discounts but does not decide.

2.3 Rural Underrepresentation & Sparse Data

This is the most structurally complex governance challenge. Rural Alabama counties face two overlapping disadvantages: lower social media activity (volume sparsity) and a model trained predominantly on urban, Reddit-sourced data (representational bias). The 2025 prototype explicitly deferred geospatial analysis due to privacy concerns ,this governance layer addresses both the sparsity and bias dimensions directly:

- **Tiered thresholds:** Rural counties use n=15 minimum, routed to a dedicated sparse-data queue with explicit uncertainty labeling rather than being silenced.

- **Transparency over false confidence:** The reviewer dashboard displays a rural_bias_flag with a plain-language note: *"Model trained on urban data — crisis language patterns in this region may be underdetected. Apply additional judgment."* Fixing the underlying bias requires responsible retraining data collection and is documented as a GSoC Phase 2 roadmap item.
- **Equity audit:** Quarterly comparison of escalation rates and response times across rural vs. urban counties detects systematic under-flagging patterns over time.

2.4 Escalation Thresholds & Human-in-the-Loop Process

CSS	Tier	Reviewer	Time Limit	Human Action Required
≥ 0.75	🔴 HIGH	Senior analyst + State BH Office	Immediate	Confirm or dismiss with documented rationale
0.50–0.74	🟡MODERATE	On-call analyst	4 hours	Review signal, flags, de-identified post sample
0.30–0.49	🟢 LOW	Any analyst	Next daily review	Log assessment, no immediate action
< 0.30	⚪ NOISE	System only	N/A	Archived for longitudinal trend analysis

No CSS value, regardless of magnitude, autonomously triggers resource deployment or community contact. The reviewer is always presented with the full evidence: what drove the score, confidence level, active flags, and sample size , so their judgment is genuinely informed, not procedural.

2.5 Audit Logging Structure

Every CSS computation generates an immutable audit record capturing four categories of information:

Category	What is Recorded
What the system saw	County FIPS, window, post count, S/V/G scores, active flags
What the system decided	CSS score, escalation tier, confidence estimate
What the human did	Analyst ID, action (Confirmed / Dismissed / Deferred), rationale
What happened after	Outcome confirmed, intervention deployed — logged 30 days post-event

The fourth category — outcome tracking is what transforms monitoring into a learning system. Analyst overrides and confirmed outcomes feed back into threshold calibration over time. Logs are append-only, tamper-evident, retained for 7 years per public health data governance standards, and accessible only to authorized ISSR and State BH Office personnel.

3. Governance Reflection

3.1 Primary Risk of Premature Deployment

The most serious risk of deploying this system before it is ready is not a technical failure , it is an institutional one. Premature deployment creates two parallel failure modes with asymmetric harm profiles. A false positive flagging a county not in genuine crisis wastes scarce behavioral health resources and, when it recurs, erodes institutional trust in AI-assisted public health tools. A false negative missing a genuine crisis in a rural county whose distress language the BERT model was never trained to recognize means people who needed intervention did not receive it. The false negative is the more ethically serious failure. But the false positive is what gets the system switched off which makes the false negative permanent.

There is also a compounding institutional risk: a high-profile failure does not just affect AI4MH. It makes every future AI public health tool harder to deploy, because trust once lost in a sensitive domain is slow to recover. The prototype-to-production transition must therefore include prospective validation against ground-truth outcomes ER admissions, crisis hotline volume, ADPH records before any operational use.

3.2 The Single Most Important Safeguard

The single most important safeguard is mandatory human confirmation before any intervention action is taken. The CSS is a decision-support signal, not a decision. But this safeguard only works if the right human is in the loop — not just any reviewer. Three conditions must be met simultaneously:

- **Verified expertise** — the reviewer is trained in behavioral health crisis patterns, not just a data analyst who happens to have dashboard access
- **Institutional authority** — they hold the power to escalate findings to county and state behavioral health offices
- **Full transparency** — they are presented with what drove the signal, confidence level, active flags, and known limitations — not just a color-coded alert

Access controls are therefore not optional. The reviewer dashboard must be role-gated so that HIGH and MODERATE alerts reach only pre-verified qualified personnel. A human in the loop who sees only a red alert with no context is not exercising judgment — they are rubber-stamping the algorithm. That is not governance. A system that routes every signal through a genuinely informed, institutionally empowered human can survive imperfect components. A system that hands that signal to the wrong person cannot.

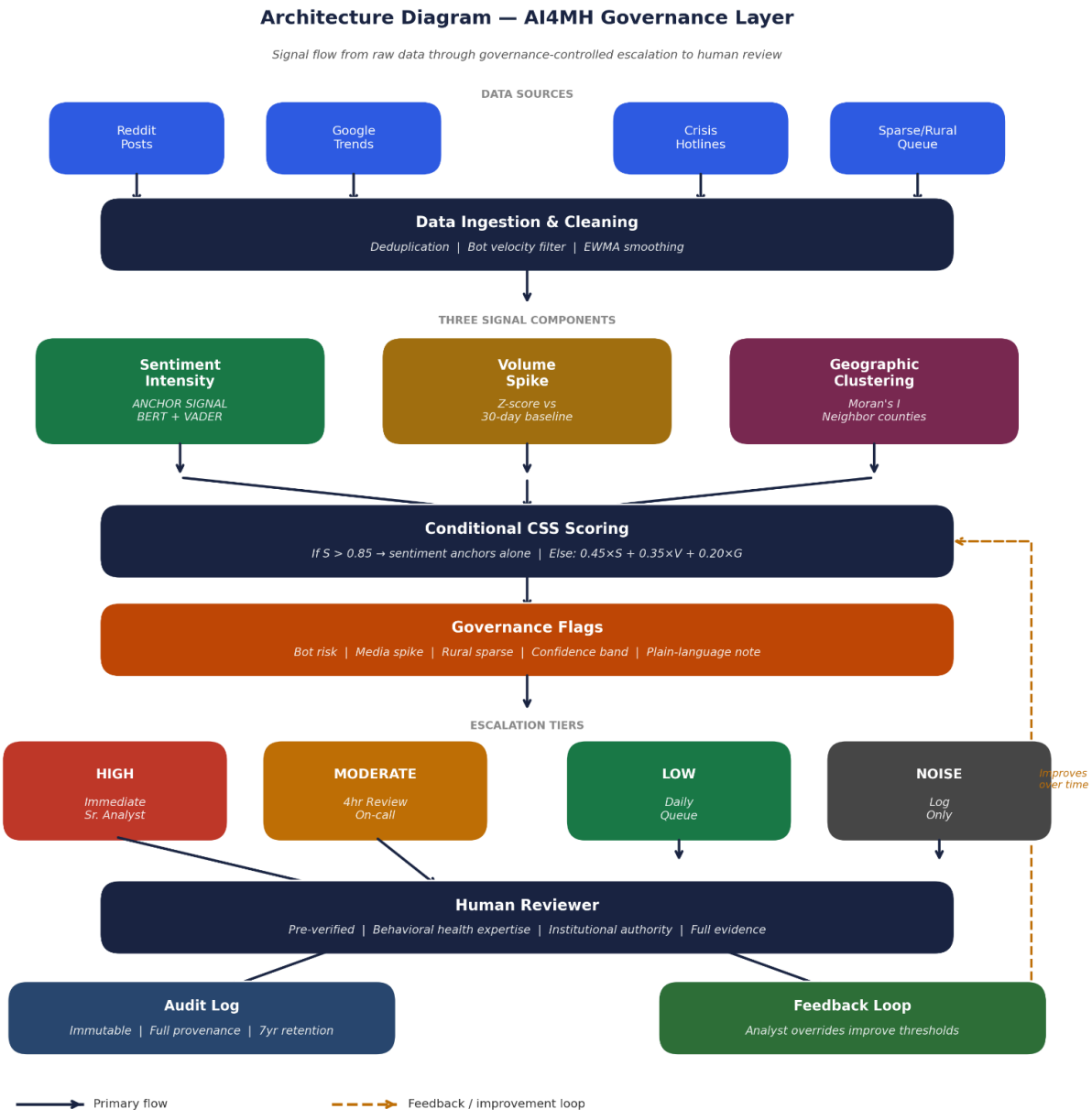

Figure 1: AI4MH Governance Layer — Signal flow from data ingestion through conditional CSS scoring, governance flagging, tiered escalation, human review, and audit-logged feedback loop. Dashed line indicates the model improvement pathway from analyst outcomes back to threshold calibration.

A Note from the Contributor

I have always been curious about mental health , I built a side project around it independently, outside of any academic requirement. Beyond GSoC, this project carries a deeper importance for me: it addresses the root of human wellbeing, and that is not something I take lightly. I genuinely want to be part of this and will commit my full capacity to developing a feasible, trustworthy solution by the end of it. Thank you for making such meaningful work open to collaboration through GSoC.

— *Swasti Jain*